Data Blueprint

Subsystem Name: GLVMS – Guest Lecture & Visitor Management System

Institution: RTMSSU
Integration Type: Central Portal + Subsystem (Role-Based)

1. Entity & Field Mapping (Core Objects & Fields)

1.1- User (Common Login – Visitor/Faculty/Security)

Field Name	Description
User_ID (VIS/FAC/SEC-XXXX)	Unique user identifier
Full_Name	Name of user
Email	Email address
Mobile_Number	Contact number
Password (Encrypted)	Login authentication
Role	Visitor / Faculty / Security / Admin
Account_Status	Active / Inactive

1.2- **Visitor Profile**

Field Name	Description
Visitor_ID	Auto-generated unique ID
Full_Name	Visitor name
Mobile_Number	Contact number
Email_Address	Email
ID_Proof_Type	Aadhar / PAN / Other
ID_Proof_Number	Identity verification
Registration_Date	Date of registration

1.3 - Visit Request (Visiting Pass Request)

Field Name	Description
Request_ID	Unique request ID
Visitor_ID	Linked visitor
Purpose_of_Visit	Guest Lecture / Meeting / Other
Date_of_Visit	Requested visit date
Department/Host	Department to visit
Request_Status	Pending / Approved / Rejected
Request_Timestamp	Time of request

1.4 - Guest Lecture Event (Faculty Module)

Field Name	Description
Lecture_ID	Unique lecture ID
Lecture_Topic	Topic of guest lecture
Speaker_Name	Guest speaker name
Department	Hosting department
Date	Lecture date
Time	Lecture timing
Venue	Lecture hall/location
Created_By (Faculty_ID)	Faculty coordinator

1.5 - QR Entry Pass (Auto Generated After Approval)

Field Name	Description
Pass_ID	Unique pass ID
Visitor_ID	Linked visitor
Request_ID	Linked request
QR_Code_Token	Unique QR for entry
Approval_Status	Approved / Rejected
Valid_Date	Allowed visit date
Pass_Generation_Time	Timestamp

1.6 - Security Verification Log (Gate Check-in)

Field Name	Description
Log_ID	Unique log ID
Pass_ID	Scanned pass reference
Visitor_ID	Visitor reference
Scan_Time	QR scan timestamp
Verification_Status	Valid / Invalid
Entry_Status	Checked-In / Exited
Security_Staff_ID	Guard verifying entry

2. Data Sensitivity Classification (Sensitive vs Public)

2.1 Highly Sensitive Data (Strictly Protected)

- Password (Encrypted)
- ID Proof Number
- QR Code Token
- Security Logs
- User Authentication Data

2.2 Sensitive Data (Restricted Access)

- Mobile Number
- Email Address
- Visitor Details
- Visit Requests
- Entry & Exit Records

2.3 Public / Low Sensitivity Data

- Lecture Topic
- Department Name
- Venue
- Visit Purpose (General)
- Request Status (Approved/Rejected)

2.4 Security Protocol:

- Encryption for passwords
- Role-Based Access Control (RBAC)
- Restricted database access for sensitive fields

3. Operations (Read / Create / Update Permissions)

3.1 Visitor Role

Operation	Access
Create	Register profile & request visiting pass
Read	View request status & QR pass
Update	Limited profile update
Delete	Not Allowed

3.2 Faculty Role

Operation	Access
Read	View all visitor requests
Update	Approve/Reject visitor requests
Create	Manage guest lecture events
Delete	Limited (lecture management only)

3.3 Security Role

Operation	Access
Read	View QR pass details
Create	Generate entry verification logs (on scan)
Update	Not Allowed (logs are immutable)
Delete	Not Allowed (audit compliance)

3.4 Central Portal API Access (Integration Scope)

Data	Access Type
Visitor Basic Info	Read
Visit Request Status	Read
Guest Lecture Events	Read
Daily Visitor Analytics	Read
Security Logs	Restricted Read (if required)

4. Data Storage Strategy (Local vs Core Portal)

4.1 Data Stored Locally (GLVMS Database)

- User credentials (encrypted)
- ID proof details
- QR pass tokens
- Security verification logs
- Detailed visitor records

4.2 Data Managed via Central Portal APIs

- Visitor ID & basic profile
- Visit approval status
- Guest lecture schedule
- Visitor attendance/entry confirmation

5. System Workflow

- Single unified login for Visitor, Faculty, and Security
- Visitor registers and requests visiting pass
- Faculty dashboard displays pending requests
- Faculty approves/rejects requests
- On approval, QR pass is auto-generated
- Security verifies QR at entry gate
- Faculty manages guest lecture events in the same subsystem